



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

First Semester M.Tech. (VDE) (Credit System) Degree Examinations

December 2024 – January 2025

22VDE102 – EMBEDDED SYSTEM DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

	Unit – I	Marks	BT*	CO*	PO*
1. a) Explain the following:					
i. Propagation Delay					
ii. Transport delay					
iii. Inertial Delay					
iv. Rise Time					
v. Fall Time		10	L*2	1	3
b) Define Static-0 hazard and static-1 hazard and Explain with neat digital circuit and Verilog module.		10	L3	1	3
2. a) With the help of logic circuit, explain the structural faults of an embedded system.		10	L2	1	3
b) Explain first order RC circuit model with necessary equations and Plot the Effect of the Parasitic Components on Signal Rise Time.		10	L2	1	3
	Unit – II				
3. a) What is UML diagram? List the common UML diagrams. Explain the use case diagram for a simple data acquisition diagram.		10	L2	2	3
b) What are the significance of fork and Join used in UML? Draw the sequence diagram for making, converting and displaying the time interval measurement in a simple counter design.		10	L2	2	3
4. a) What is dynamic modeling with UML? List the important elements of a dynamic model. Explain the different interaction diagrams used in modeling the dynamic behavior of the system.		10	L2	2	3
b) Explain the following in UML:					
i) Guard Conditions					
ii) Composite states		10	L2	2	3
	Unit – III				
5. a) Explain the following with the help of diagram:					
i) Water fall model					
ii) V cycle model					
What is the difference between the two models?		10	L2	3	3
b) Explain the system requirements specification for a digital counter. The counter is able to measure frequency, period, time interval and events. Draw the front panel and use cases.		10	L2	3	3
6. a) Briefly explain the Co-design process with necessary diagrams.		10	L2	3	3
b) What is the purpose and goal of a functional design? What are the steps that need to be considered while starting the function design of a system? Explain the first level decomposition of a simple input output task.		10	L2	3	3
	Unit – IV				
7. a) What kind of information should be recorded and kept when sudden failure occurs during debugging? Explain with neat sketches and current equations considering Thevenin's model as example.		10	L2	4	3

b)	What are the three types of tests generally recognized when debugging and testing modules, explain with example	10	L2	4	3
8. a)	What is path sensitizing? Explain testing and debugging process, considering basic combinational logic circuit for single variable–two paths	10	L2	4	3
b)	With necessary sketches explain the following,				
i)	Masking and untestable faults				
ii)	Bridge faults	10	L2	4	3
Unit – V					
9. a)	What is an operating system? Where it is used and what are its primary functions? With neat sketches explain the basic components of an operating system and their interfaces.	10	L2	5	3
b)	What is kernel? Classify and explain the kernels based on its design.	10	L2	5	3
10. a)	Explain the interrupt handling mechanism under VxWorks kernel.	10	L2	5	3
b)	Explain the following.				
i)	Inter task communication under VxWorks				
ii)	Mutual exclusion mechanisms supported by VxWorks	10	L2	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
